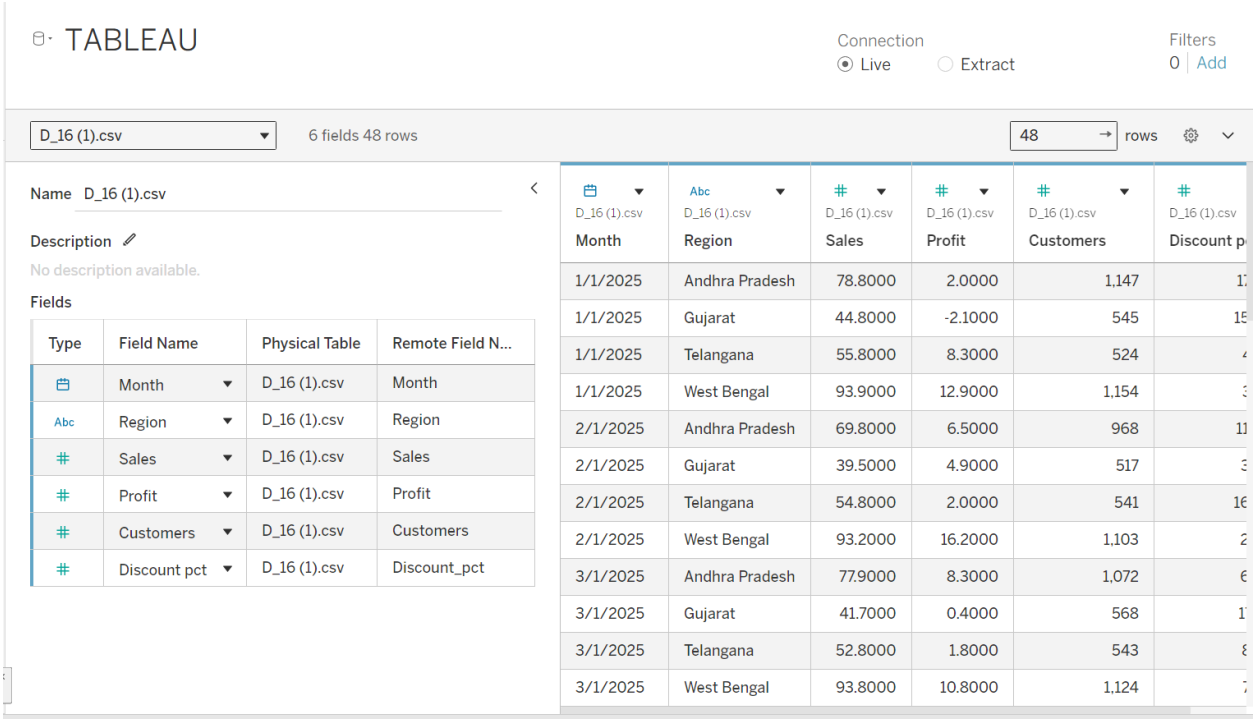
DSA06 - Data Handling and Visualization

Rg. No: 16 Reg. No: 192424097 **Name: K DEEKSHA SREE**

Data: 48 rows covering 12 months of 2025 for four regions (Andhra Pradesh, Telangana, Gujarat, West Bengal). Fields: Month, Region, Sales (Rs lakh), Profit (Rs lakh), Customers, Discount_pct.

T1. Connect and check

Connected to D_16.csv as a Text file in Tableau Public and changed Month from Text to Date.



T2. Bar chart

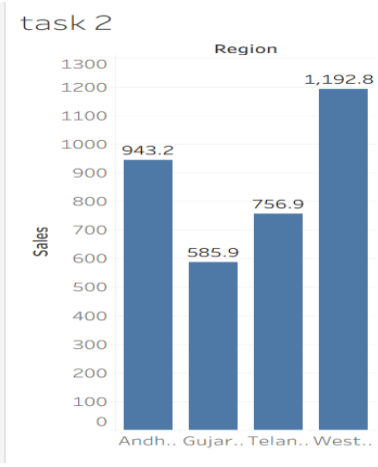


Figure 1: Sales by Region, sorted high to low, with labels

Region	Sales (Rs lakh)	Share of total	Rank
West Bengal	1,192.8	34.3%	1 (top)
Andhra Pradesh	943.2	27.1%	2
Telangana	756.9	21.8%	3
Gujarat	585.9	16.8%	4 (lowest)

Answer: Top region is West Bengal (1,192.8) and the lowest is Gujarat (585.9). West Bengal's share of total sales is 1,192.8 / 3,478.8 = 34.3%.

T3. Line chart and trend

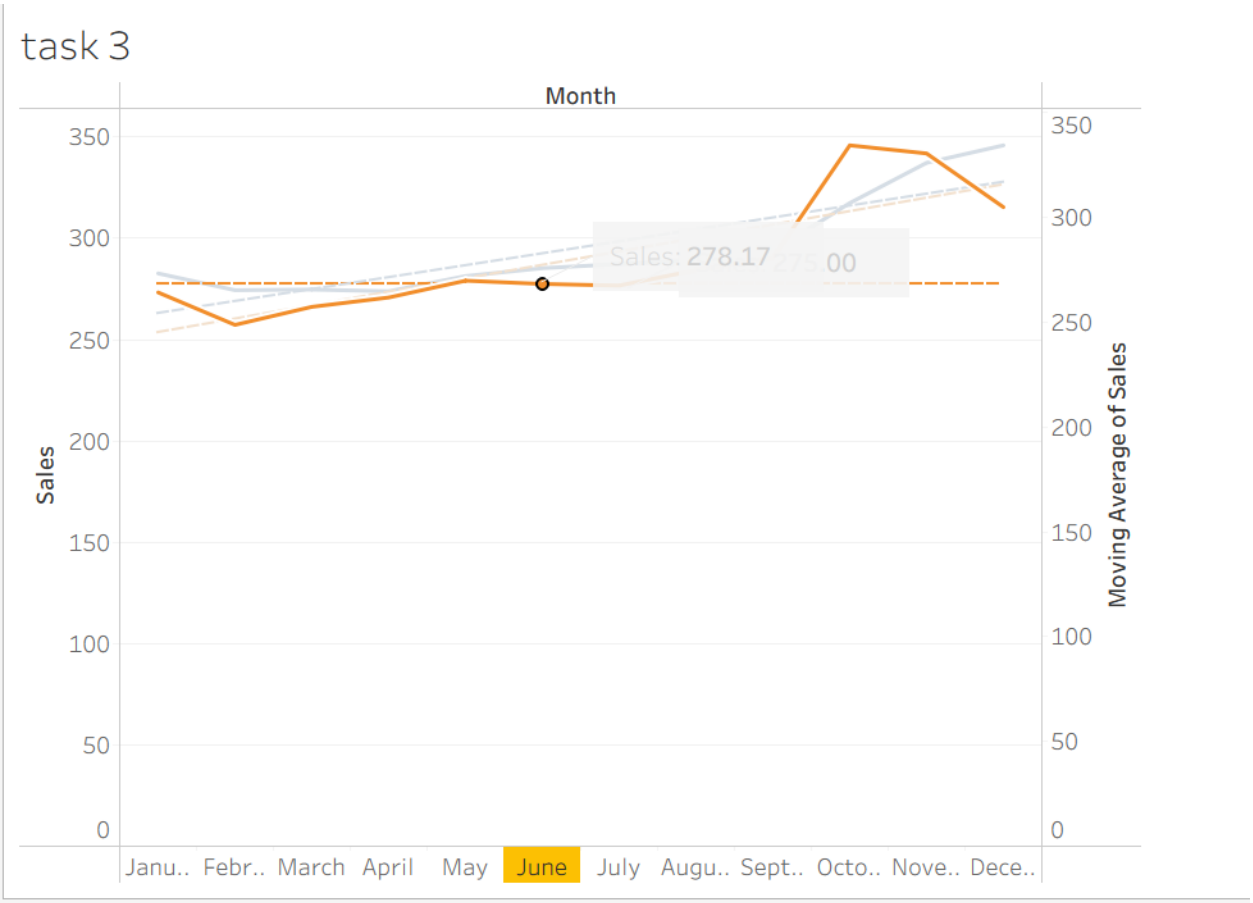


Figure 2: Total Sales by Month with linear trend line and 3-month moving average

Item	Result
Peak month and value	October 2025, 345.9 Rs lakh
3-month moving average at June	275.8 Rs lakh (average of Apr 270.8, May 279.1, Jun 277.5)
Trend slope (Describe Trend Model)	about +6.63 Rs lakh per month
R-squared / p-value	0.680 / 0.001 (significant)

Answer: Sales rose steadily through the year, with a sharp jump in October (peak 345.9), followed by a decline in November and December. The trend line shows sales growing by roughly 6.6 Rs lakh each month.

T4. Scatterplot

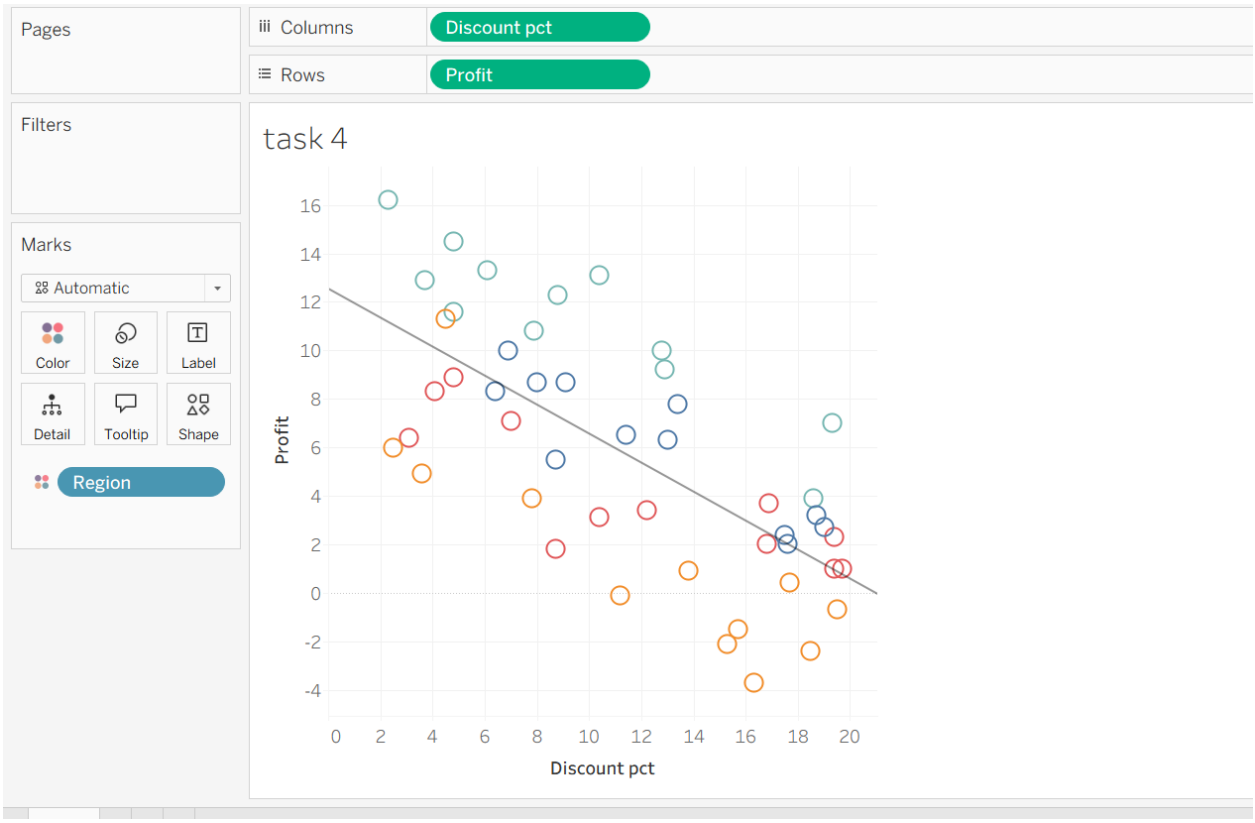
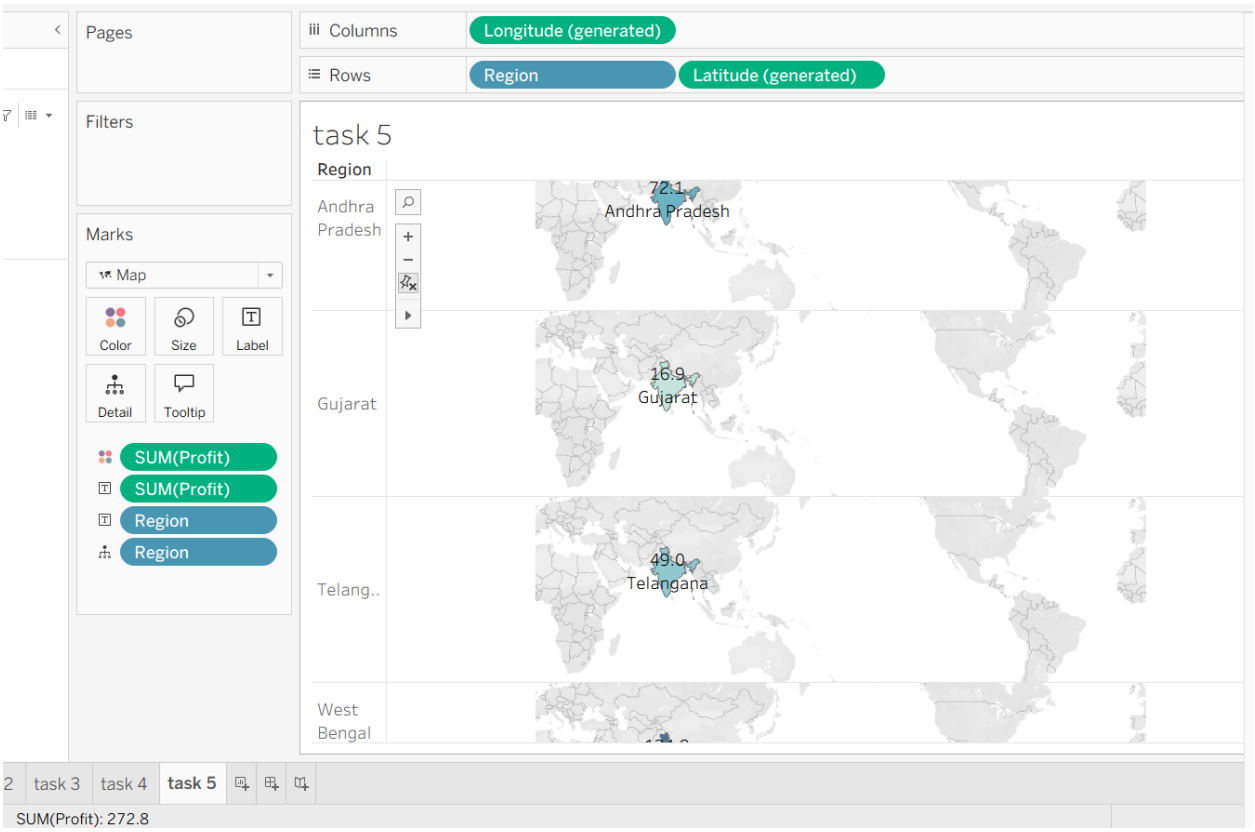


Figure 3: Discount_pct vs Profit (aggregation off), coloured by Region, one overall trend line

Item	Result
Slope	-0.598 (Profit falls about 0.6 Rs lakh per 1-point rise in discount)
R-squared	0.485
r (sign of the slope)	-0.696

Meaning for the business: Higher discounts are linked to lower profit (moderate-to-strong negative correlation), so heavy discounting is eroding margins and should be limited. Within each individual region the relationship is even stronger (r between -0.83 and -0.90).

T5. Filled map



Region	Sales	Profit	Profit margin
Gujarat (lowest profit)	585.9	16.9	2.9%
Telangana	756.9	49.0	6.5%
Andhra Pradesh	943.2	72.1	7.6%
West Bengal	1,192.8	134.8	11.3%

Answer: Gujarat has the lowest total profit (16.9 Rs lakh). Its profit margin is 16.9 / 585.9 = 2.9%.

T6. Dashboard with a filter

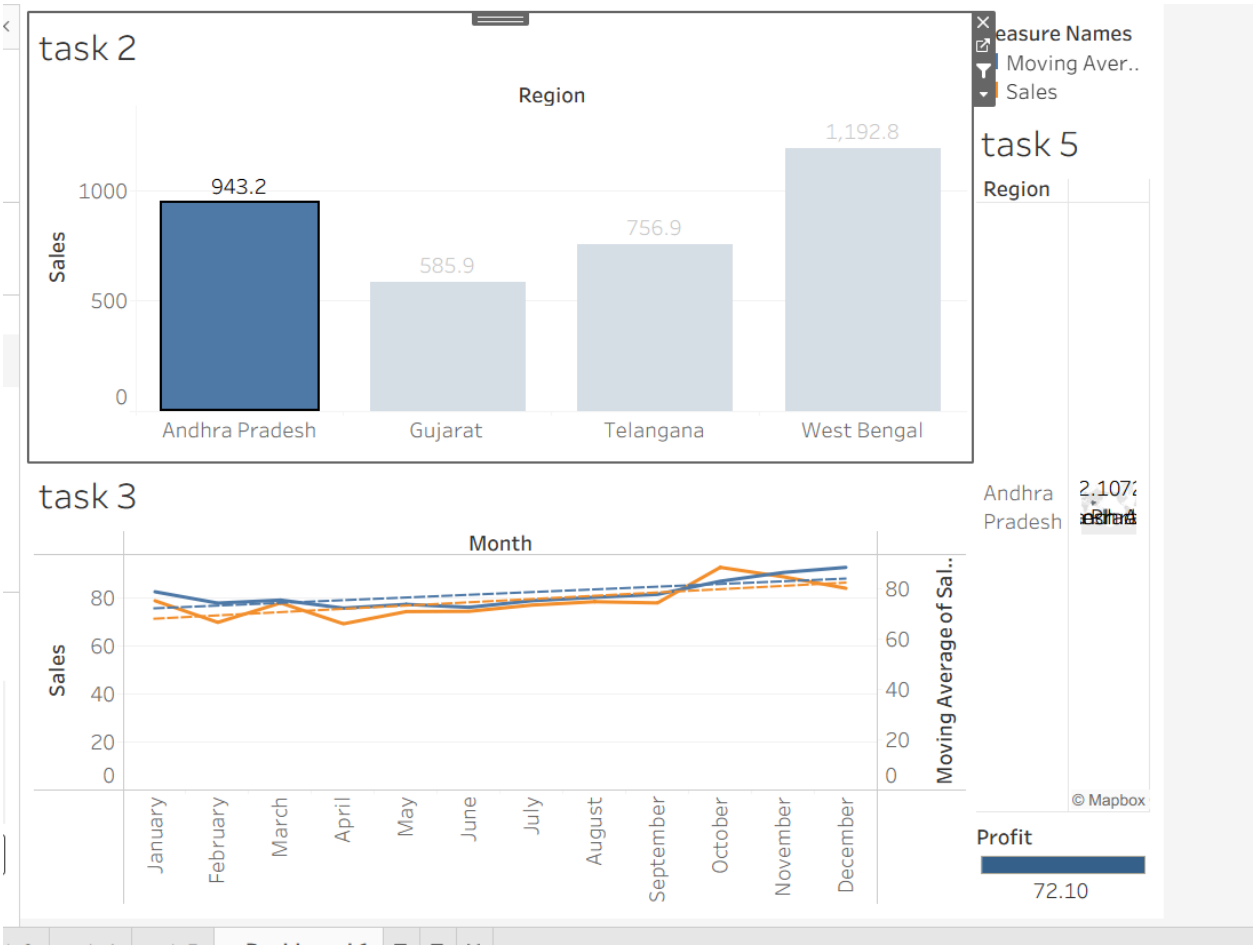


Figure 5: Dashboard with bar chart (Use as Filter), line chart and map; Andhra Pradesh selected

The bar chart was set as a filter (Use as Filter). Clicking the second-highest sales region, Andhra Pradesh, updates the other sheets:

Measure (Andhra Pradesh)	Value
Total Sales	943.2 Rs lakh
Total Profit	72.1 Rs lakh
Profit margin	72.1 / 943.2 = 7.6%

T7. Compare with R

1. Effort and setup: In Tableau the line chart, trend line and moving average are built by drag-and-drop and menu options (Analytics pane, Quick Table Calculation) with no code. In R (ggplot2) each element is written as code, for example geom_line() and geom_smooth(method = "lm"), and the moving average needs a package such as zoo.

2. Reproducibility and customisation: An R script can be re-run on new data and gives exact control over every aspect of the chart, and it can be version controlled. A Tableau chart is interactive (tooltips, filters, click to explore) but repeating it on new data means manual steps, and fine styling is limited to what the interface offers.

R-code:

```
library(ggplot2)

library(dplyr)

library(zoo)
```

```
d <- read.csv("D_16.csv")
d$Month <- as.Date(d$Month)

m <- d %>%
  group_by(Month) %>%
  summarise(Sales = sum(Sales)) %>%
  mutate(MA3 = rollmean(Sales, 3, fill = NA, align = "right"))

ggplot(m, aes(Month, Sales)) +
  geom_line(colour = "steelblue") +
  geom_point(colour = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, linetype = "dashed", colour = "grey30") +
  geom_line(aes(y = MA3), colour = "red") +
  labs(title = "Total Sales by Month",
       x = "Month (2025)", y = "Total Sales (Rs lakh)")
```